Assignment 1

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Part 1: Simulating the Forward Diffusion

Derivations

Throughout, X_t solves the Ornstein–Uhlenbeck SDE

$$\mathrm{d}X_t = -\theta X_t\,\mathrm{d}t + \sqrt{2\theta}\,\mathrm{d}W_t, \qquad \theta > 0, \tag{1}$$

with initial density $p_0(x) = \frac{1}{2}\varphi(x \mid -3, 0.5^2) + \frac{1}{2}\varphi(x \mid 3, 0.5^2)$. We use the Itô rules $(\mathrm{d}t)^2 = 0$, $\mathrm{d}t\,\mathrm{d}W_t = 0$, $(\mathrm{d}W_t)^2 = \mathrm{d}t$, and $\mathbb{E}[\mathrm{d}W_t] = 0$ (since $\mathrm{d}W_t \sim \mathcal{N}(0, \mathrm{d}t)$ and is independent of X_t).

1(a) Stationary distribution

Mean. Taking expectations of (1),

$$\mathrm{d}\,\mathbb{E}[X_t] = -\theta\,\mathbb{E}[X_t]\,\mathrm{d}t + \sqrt{2\theta}\,\mathbb{E}[\mathrm{d}W_t] \quad \Longrightarrow \quad \frac{\mathrm{d}m}{\mathrm{d}t} = -\theta\, m(t).$$

Separating variables, $\int \frac{\mathrm{d}m}{m} = -\theta \int \mathrm{d}t$, so $\ln|m| = -\theta t + c$, i.e. $m(t) = Ce^{-\theta t}$. Setting $t = 0$ gives $C = m(0)$, hence

$$m(t) = m(0)\, e^{-\theta t} \xrightarrow[t\to\infty]{} 0.$$

Second moment. Apply Itô's lemma, $\mathrm{d}f = \partial_t f\,\mathrm{d}t + \partial_x f\,\mathrm{d}X_t + \frac{1}{2}\partial_{xx} f\,(\mathrm{d}X_t)^2$, to $f(x) = x^2$, for which $\partial_t f = 0$, $\partial_x f = 2x$, $\partial_{xx} f = 2$:

$$\mathrm{d}(X_t^2) = 2X_t\,\mathrm{d}X_t + (\mathrm{d}X_t)^2. \tag{2}$$

Squaring (1),

$$(\mathrm{d}X_t)^2 = \theta^2 X_t^2 (\mathrm{d}t)^2 + 2\theta(\mathrm{d}W_t)^2 - 2\theta\sqrt{2\theta}\, X_t\,\mathrm{d}t\,\mathrm{d}W_t = 2\theta\,\mathrm{d}t.$$

Substituting this and (1) into (2),

$$\mathrm{d}(X_t^2) = 2X_t\big(-\theta X_t\,\mathrm{d}t + \sqrt{2\theta}\,\mathrm{d}W_t\big) + 2\theta\,\mathrm{d}t = \big(-2\theta X_t^2 + 2\theta\big)\mathrm{d}t + 2\sqrt{2\theta}\, X_t\,\mathrm{d}W_t.$$

Taking expectations (the Itô integral term has zero mean),

$$\frac{\mathrm{d}\,\mathbb{E}[X_t^2]}{\mathrm{d}t} = -2\theta\,\mathbb{E}[X_t^2] + 2\theta. \tag{3}$$

Variance. By definition $\Sigma(t) = \mathbb{E}[X_t^2] - m(t)^2$, so

$$\frac{\mathrm{d}\Sigma}{\mathrm{d}t} = \frac{\mathrm{d}\,\mathbb{E}[X_t^2]}{\mathrm{d}t} - 2m(t)\frac{\mathrm{d}m}{\mathrm{d}t}.$$

Using (3) and $\mathrm{d}m/\mathrm{d}t = -\theta m$,

$$\frac{\mathrm{d}\Sigma}{\mathrm{d}t} = -2\theta\,\mathbb{E}[X_t^2] + 2\theta + 2\theta\, m(t)^2 = -2\theta\big(\mathbb{E}[X_t^2] - m(t)^2\big) + 2\theta = -2\theta\,\Sigma(t) + 2\theta.$$

Separating variables, $\int_0^t \frac{\mathrm{d}\Sigma}{\Sigma - 1} = -2\theta \int_0^t \mathrm{d}s$, so $\ln\left|\frac{\Sigma(t)-1}{\Sigma(0)-1}\right| = -2\theta t$ and

$$\Sigma(t) = 1 + \big(\Sigma(0) - 1\big)e^{-2\theta t} \xrightarrow[t\to\infty]{} 1.$$

Stationary law. The limiting mean and variance are 0 and 1. To identify the whole distribution (not only its first two moments) we use the solution derived in 1(b): conditionally on X_0, $X_t \sim \mathcal{N}(e^{-\theta t}X_0,\, 1 - e^{-2\theta t})$, whose characteristic function is $\mathbb{E}[e^{iuX_t} \mid X_0] = \exp\!\big(iue^{-\theta t}X_0 - \tfrac{1}{2}u^2(1 - e^{-2\theta t})\big)$. As $t \to \infty$ this tends to $e^{-u^2/2}$ for every X_0, and by dominated convergence $\mathbb{E}[e^{iuX_t}] \to e^{-u^2/2}$ for any initial law. Hence

$$\boxed{\pi = \mathcal{N}(0, 1).}$$

Equivalently, $\mathcal{N}(0, 1)$ is stationary: if $X_0 \sim \mathcal{N}(0, 1)$ then X_t is Gaussian with $m(t) = 0$ and $\Sigma(t) = 1$ for all t.

1(b) Solution by integrating factor

Write (1) as $\mathrm{d}X_t + \theta X_t\,\mathrm{d}t = \sqrt{2\theta}\,\mathrm{d}W_t$, which has the linear form $\dot{x} + P(t)x = Q(t)$ with $P(t) = \theta$. The integrating factor is $I_t = \exp\big(\int \theta\,\mathrm{d}t\big) = e^{\theta t}$. Define $Y_t = f(t, X_t) = e^{\theta t}X_t$. The partial derivatives are

$$\partial_t f = \theta e^{\theta t}x, \qquad \partial_x f = e^{\theta t}, \qquad \partial_{xx} f = 0, \qquad \partial_{tx} f = \theta e^{\theta t}, \qquad \partial_{tt} f = \theta^2 e^{\theta t}x.$$

The multivariate Itô expansion is

$$\mathrm{d}Y_t = \partial_t f\,\mathrm{d}t + \partial_x f\,\mathrm{d}X_t + \tfrac{1}{2}\partial_{xx} f\,(\mathrm{d}X_t)^2 + \partial_{tx} f\,\mathrm{d}t\,\mathrm{d}X_t + \tfrac{1}{2}\partial_{tt} f\,(\mathrm{d}t)^2.$$

Here $(\mathrm{d}t)^2 = 0$, $\mathrm{d}t\,\mathrm{d}X_t = -\theta X_t(\mathrm{d}t)^2 + \sqrt{2\theta}\,\mathrm{d}t\,\mathrm{d}W_t = 0$, and the $(\mathrm{d}X_t)^2 = 2\theta\,\mathrm{d}t$ term is multiplied by $\partial_{xx} f = 0$. Therefore

$$\mathrm{d}(e^{\theta t}X_t) = \theta e^{\theta t}X_t\,\mathrm{d}t + e^{\theta t}\big(-\theta X_t\,\mathrm{d}t + \sqrt{2\theta}\,\mathrm{d}W_t\big) = \sqrt{2\theta}\,e^{\theta t}\,\mathrm{d}W_t.$$

Integrating from 0 to t,

$$e^{\theta t}X_t - X_0 = \sqrt{2\theta}\int_0^t e^{\theta s}\,\mathrm{d}W_s \quad\Longrightarrow\quad X_t = e^{-\theta t}X_0 + \sqrt{2\theta}\int_0^t e^{-\theta(t-s)}\,\mathrm{d}W_s.$$

Gaussianity. The stochastic integral has a deterministic integrand, so it is a limit of sums $\sum_k g(s_k)(W_{s_{k+1}} - W_{s_k})$ of independent Gaussians and is therefore Gaussian. It is independent of X_0, so $X_t \mid X_0$ is Gaussian and is fixed by its mean and variance.

Mean. Since $\mathbb{E}\big[\int_0^t g(s)\,\mathrm{d}W_s\big] = 0$,

$$\mathbb{E}[X_t \mid X_0] = e^{-\theta t}X_0 + \sqrt{2\theta}\int_0^t e^{-\theta(t-s)}\,\mathbb{E}[\mathrm{d}W_s] = e^{-\theta t}X_0.$$

Variance. $X_t - \mathbb{E}[X_t \mid X_0] = \sqrt{2\theta}\int_0^t e^{-\theta(t-s)}\,\mathrm{d}W_s$. By Itô's isometry, $\mathbb{E}\big[(\int_0^t g(s)\,\mathrm{d}W_s)^2\big] = \int_0^t g(s)^2\,\mathrm{d}s$, so

$$\mathrm{Var}(X_t \mid X_0) = 2\theta\int_0^t e^{-2\theta(t-s)}\,\mathrm{d}s = 2\theta\,e^{-2\theta t}\left[\frac{e^{2\theta s}}{2\theta}\right]_0^t = e^{-2\theta t}\big(e^{2\theta t} - 1\big) = 1 - e^{-2\theta t}.$$

Hence

$$\boxed{X_t \mid X_0 \sim \mathcal{N}\Big(e^{-\theta t}X_0,\; 1 - e^{-2\theta t}\Big).}$$

1(c) Law of X_t

From 1(b) we may write $X_t = e^{-\theta t}X_0 + \xi_t$ with $\xi_t \sim \mathcal{N}(0, 1-e^{-2\theta t})$ independent of X_0. Introduce a latent label $Z \in \{1, 2\}$ with $P(Z = 1) = P(Z = 2) = \frac{1}{2}$ such that

$$X_0 \mid Z = 1 \sim \mathcal{N}(-3, 0.5^2), \qquad X_0 \mid Z = 2 \sim \mathcal{N}(3, 0.5^2),$$

which reproduces p_0. By the law of total probability, $p_t(x) = \frac{1}{2}\, p(x \mid Z = 1) + \frac{1}{2}\, p(x \mid Z = 2)$. Conditionally on Z, X_t is a sum of two independent Gaussians, hence Gaussian, with

$$\mathbb{E}[X_t \mid Z = 1] = e^{-\theta t}\mathbb{E}[X_0 \mid Z = 1] + \mathbb{E}[\xi_t] = -3e^{-\theta t}, \qquad \mathbb{E}[X_t \mid Z = 2] = 3e^{-\theta t},$$

and, using $\mathrm{Var}(cY) = c^2\,\mathrm{Var}(Y)$ and independence,

$$\mathrm{Var}(X_t \mid Z = k) = e^{-2\theta t}\,\mathrm{Var}(X_0 \mid Z = k) + \mathrm{Var}(\xi_t) = 0.25\, e^{-2\theta t} + 1 - e^{-2\theta t} = 1 - 0.75\, e^{-2\theta t} =: v_t,$$

for $k = 1, 2$. Therefore

$$\boxed{p_t(x) = \tfrac{1}{2}\varphi\Big(x \mid -3e^{-\theta t}, v_t\Big) + \tfrac{1}{2}\varphi\Big(x \mid 3e^{-\theta t}, v_t\Big), \qquad v_t = 1 - 0.75\, e^{-2\theta t}.}$$

Checks. At $t = 0$: $v_0 = 1 - 0.75 = 0.25 = 0.5^2$, and the means are ∓ 3, recovering p_0. As $t \to \infty$: $v_t = 1 - 0.75 \lim_{t\to\infty} e^{-2\theta t} = 1$ and both means tend to 0, so $p_t \to \varphi(x \mid 0, 1) = \pi$, consistent with 1(a).

1(d) Dependence on θt only

Set $s = \theta t$. Then

$$p_t(x) = \tfrac{1}{2}\varphi\big(x \mid -3e^{-s},\, 1 - 0.75e^{-2s}\big) + \tfrac{1}{2}\varphi\big(x \mid 3e^{-s},\, 1 - 0.75e^{-2s}\big)\,.$$

Neither θ nor t appears anywhere except through s, so the law of X_t depends on (θ, t) only through the product θt.

This also follows directly from the SDE. Let $\tilde{X}_s = X_{s/\theta}$. Since $\tilde{W}_s = \sqrt{\theta}\, W_{s/\theta}$ is a standard Brownian motion,

$$\mathrm{d}\tilde{X}_s = -\theta \tilde{X}_s \frac{\mathrm{d}s}{\theta} + \sqrt{2\theta}\,\frac{\mathrm{d}\tilde{W}_s}{\sqrt{\theta}} = -\tilde{X}_s\,\mathrm{d}s + \sqrt{2}\,\mathrm{d}\tilde{W}_s,$$

which is the $\theta = 1$ OU process with the same initial law. Changing θ is therefore only a rescaling of time.

Implementation

Equation (1) is discretised with the Euler–Maruyama scheme

$$X_{t+\Delta t} = X_t - \theta X_t\,\Delta t + \sqrt{2\theta\Delta t}\, Z_t, \qquad Z_t \overset{\text{iid}}{\sim} \mathcal{N}(0, 1),$$

using the defaults $\theta = 1$, $N = 2 \times 10^4$, $\Delta t = 10^{-3}$ and a fixed random seed. The horizon is $T = 5.40$, the value obtained in Part 2(d). The code and instructions to reproduce every figure are in the `Part 1 code` folder.

(a) Sample paths

Figure 1 shows ten trajectories (simulated with a separate seed, so that the figure shows typical paths rather than a rare large excursion). Paths start near ± 3, are pulled towards 0 on a time scale of order $1/\theta$, and then fluctuate with unit variance, mixing between the two former modes.

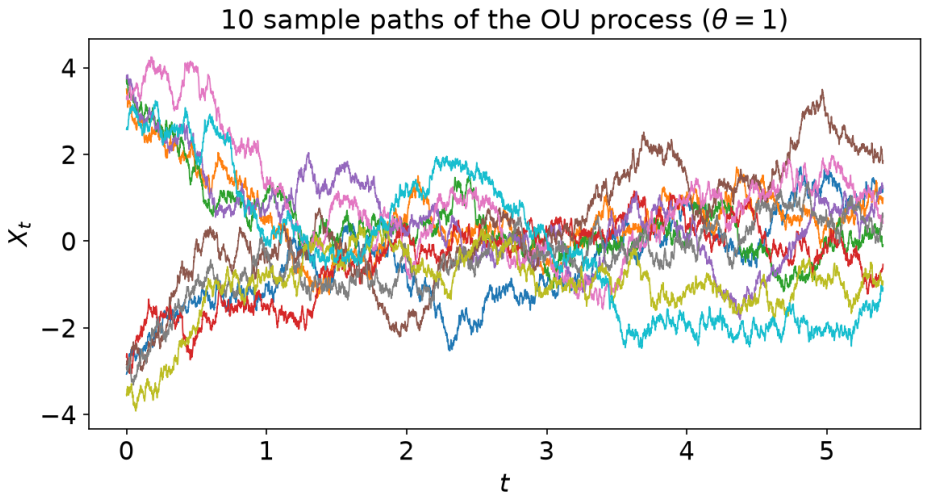


Figure 1: Ten Euler–Maruyama sample paths of the OU process, $\theta = 1$.

(b) Empirical histograms versus the exact density

Figure 2 overlays the histogram of the N simulated particles (100 bins on $[-5, 5]$) on the exact density p_t from 1(c) at five equally spaced times in $[0, T]$. We measure the discrepancy as $\max_j |\hat{p}_t(x_j) - p_t(x_j)|$ over bin centres x_j.

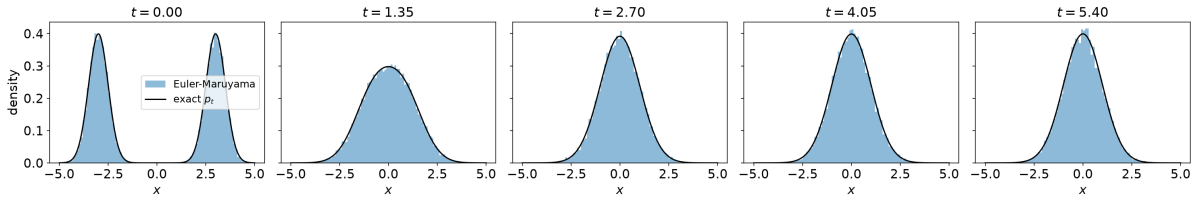


Figure 2: Simulated histograms (shaded) against the exact density p_t (black).

t	0.00	1.35	2.70	4.05	5.40
$\max_j \lvert\hat{p}_t - p_t\rvert$	0.0244	0.0149	0.0210	0.0248	0.0295

Table 1: Largest pointwise histogram error at each snapshot time.

The largest discrepancy observed is 0.030. This error is about what Monte Carlo noise alone predicts: with bin width $h = 0.1$ and $N = 2 \times 10^4$, the standard error of a bin height near a peak of height ≈ 0.4 is $\sqrt{p/(Nh)} \approx 0.014$, and the maximum over 100 bins is a few such standard errors. The error does not grow with t, so the time-discretisation bias is negligible at $\Delta t = 10^{-3}$.

(c) Disappearance of bimodality

Under the dynamics both mixture components move towards the origin, with means $\pm 3e^{-\theta t}$, while their variance v_t grows from 0.25 to 1. The two bumps therefore approach and widen until

they merge into a single peak (Figure 3).

For a symmetric mixture $\frac{1}{2}\varphi(x \mid -\mu, v) + \frac{1}{2}\varphi(x \mid \mu, v)$, the origin is always a critical point, and

$$p''(0) \propto \frac{\mu^2}{v} - 1.$$

Hence the density is bimodal (with a local minimum at 0) iff $\mu^2 > v$, and unimodal otherwise. With $\mu = 3e^{-\theta t}$ and $v_t = 1 - 0.75e^{-2\theta t}$, the condition $\mu^2 = v_t$ gives $9.75\, e^{-2\theta t} = 1$, i.e.

$$t^* = \frac{\ln 9.75}{2\theta} \approx 1.1386 \quad (\theta = 1).$$

Numerical check. Counting local maxima of the exact p_t on a grid of 4001 points in $[-6, 6]$, stepping t by 10^{-4}, gives a transition at $t = 1.1387$, matching t^*.

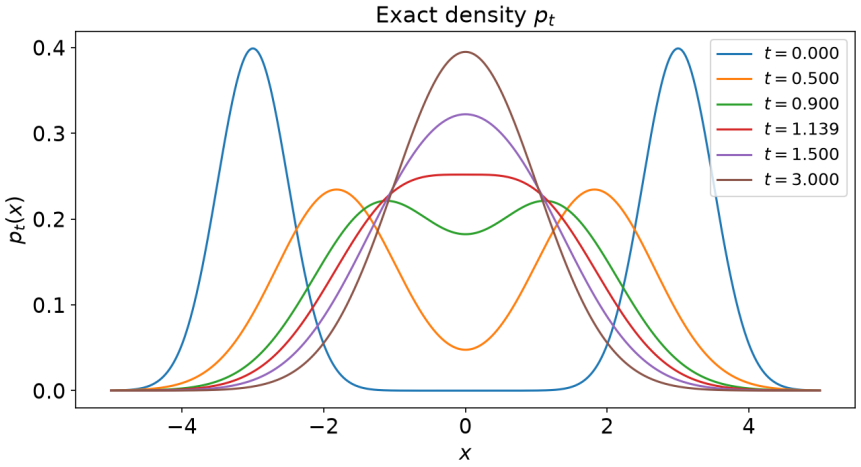


Figure 3: Exact p_t at several times, including t^*.

(d) Collapse under rescaled time

We simulate $\theta \in \{0.5, 1, 2\}$ and take snapshots at $t = s/\theta$ for $s = \theta t \in \{0, 1.35, 2.70, 4.05, 5.40\}$. Figure 4 shows that the three histograms coincide at each s and match the exact density evaluated at $\theta t = s$. All pointwise errors lie between 0.015 and 0.031, the same Monte Carlo noise level as in (b). This confirms derivation 1(d): the law of X_t depends on θ and t only through θt.

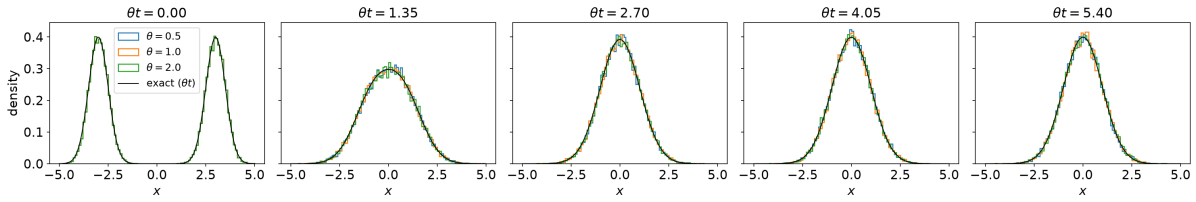


Figure 4: Histograms for $\theta \in \{0.5, 1, 2\}$ at equal rescaled times θt, with the exact density.

Part 2: Convergence to Equilibrium in Wasserstein Distance

The one-dimensional W_2 estimator

For probability measures μ, ν on $\mathbb{R}$ with finite second moments, $W_2^2(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu,\nu)} \mathbb{E}_{(X,Y)\sim\gamma}|X - Y|^2$, where $\Gamma(\mu, \nu)$ is the set of couplings. Given samples $x_1, \ldots, x_n \sim \mu$ and $y_1, \ldots, y_n \sim \nu$, we

use

$$\widehat{W}_2^2 = \frac{1}{n}\sum_{i=1}^{n}\big(x_{(i)} - y_{(i)}\big)^2,$$

where $x_{(i)}$ and $y_{(i)}$ are the order statistics.

Why sorting is optimal. Between two sets of n equally weighted points, an optimal coupling can be taken to be a one-to-one matching of the x_i with the y_j. Suppose two matched pairs cross, i.e. $x_a < x_b$ but $y_a > y_b$. Then

$$(x_a - y_a)^2 + (x_b - y_b)^2 - \big[(x_a - y_b)^2 + (x_b - y_a)^2\big] = 2(x_b - x_a)(y_a - y_b) > 0,$$

so swapping the partners strictly lowers the cost. Hence an optimal matching has no crossings, which means it pairs the i-th smallest x with the i-th smallest y. So $\widehat{W}_2$ is exactly the W_2 distance between the two sample sets, and it converges to $W_2(\mu,\nu)$ as $n \to \infty$. Because both samples are random, $\widehat{W}_2$ does not reach 0 even for identical distributions; it levels off at a small noise floor that shrinks as n grows.

2(c) Proof of the contraction bound by synchronous coupling

Let $\gamma^\star$ be an optimal coupling of p_0 and π (it exists since both have finite second moments), and draw $(X_0, Y_0) \sim \gamma^\star$, so that $\mathbb{E}|X_0 - Y_0|^2 = W_2^2(p_0, \pi)$. Drive two copies of (1) with the *same* Brownian motion W_t:

$$\mathrm{d}X_t = -\theta X_t\,\mathrm{d}t + \sqrt{2\theta}\,\mathrm{d}W_t, \qquad \mathrm{d}Y_t = -\theta Y_t\,\mathrm{d}t + \sqrt{2\theta}\,\mathrm{d}W_t.$$

Subtracting, the noise cancels exactly and $D_t = X_t - Y_t$ satisfies the deterministic ODE

$$\mathrm{d}D_t = -\theta D_t\,\mathrm{d}t \quad\Longrightarrow\quad D_t = e^{-\theta t} D_0.$$

Now $X_t \sim p_t$. Since π is stationary (1(a)) and $Y_0 \sim \pi$, also $Y_t \sim \pi$ for all t. Thus (X_t, Y_t) is a coupling of p_t and π, and by the definition of W_2 as an infimum over couplings,

$$W_2^2(p_t, \pi) \le \mathbb{E}|X_t - Y_t|^2 = e^{-2\theta t}\,\mathbb{E}|X_0 - Y_0|^2 = e^{-2\theta t}\,W_2^2(p_0, \pi).$$

Taking square roots gives

$$\boxed{W_2(p_t, \pi) \le e^{-\theta t}\,W_2(p_0, \pi).}$$

2(d) Choice of the horizon T

The bound guarantees $W_2(p_T, \pi) \le \varepsilon$ as soon as $e^{-\theta T} W_2(p_0, \pi) \le \varepsilon$, so the smallest such time is

$$T = \frac{1}{\theta}\ln\frac{W_2(p_0, \pi)}{\varepsilon}, \qquad \varepsilon = 10^{-2}.$$

The sorted-sample estimate with $N = 2 \times 10^4$ gives $W_2(p_0, \pi) = 2.2106$, so with $\theta = 1$ we get $T = 5.3984$. We round up and use

$$\boxed{T = 5.40}$$

for the rest of the assignment, including the Part 1 figures.

Implementation

All distances below use the sorted-sample estimator with $N = 2 \times 10^4$ samples on each side. Samples of p_t are the Euler–Maruyama particles from Part 1 ($\theta = 1$, $\Delta t = 10^{-3}$). They are compared with a fixed set of fresh samples from π and from p_0, drawn independently of the simulation. Distances are computed at 55 equally spaced times in $[0, T]$.

2(a) Distance to equilibrium and to the start

Figure 5 shows $W_2(p_t, \pi)$ falling from 2.21 at $t = 0$ to about 0.02, while $W_2(p_t, p_0)$ rises from near 0 to 2.21: as the process approaches equilibrium it moves away from where it started. The curves cross at $t \approx 0.5$, where p_t is equally far from both. By $t \approx 3$ both curves are flat, so the process has essentially reached π, and the distance back to p_0 equals $W_2(p_0, \pi)$.

Neither curve reaches exactly 0. At $t = 0$, $W_2(p_t, p_0)$ is 0.18 rather than 0, and at late times $W_2(p_t, \pi)$ levels off at about 0.02. Both are the noise floor of the estimator: two independent sample sets from the same distribution are still a positive distance apart. The floor is larger for p_0 because few samples fall in the gap between its two bumps, so the sorted values in that region line up poorly.

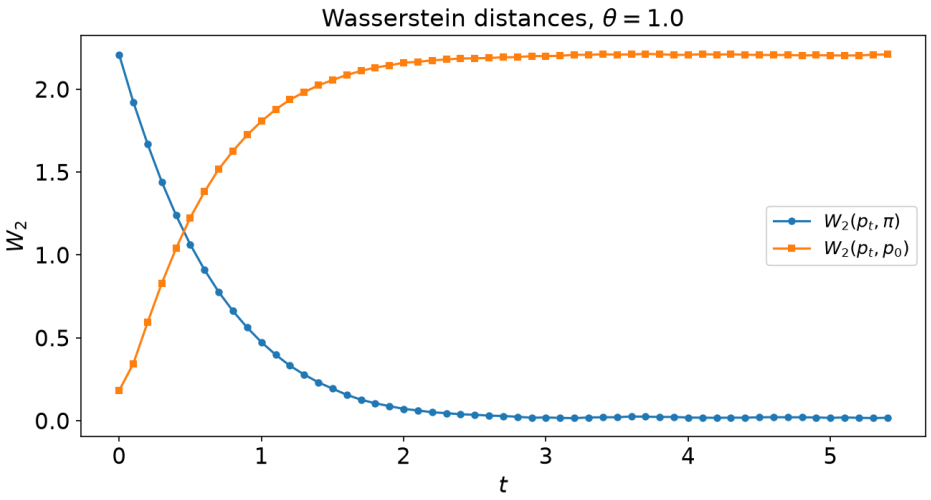


Figure 5: Estimated $W_2(p_t, \pi)$ and $W_2(p_t, p_0)$ over $[0, T]$, $\theta = 1$.

2(b) Contraction bound

The estimate of the initial distance is $W_2(p_0, \pi) = 2.2106$. Figure 6 compares the bound $e^{-\theta t} W_2(p_0, \pi)$ with the estimated $W_2(p_t, \pi)$, and Table 2 lists selected values.

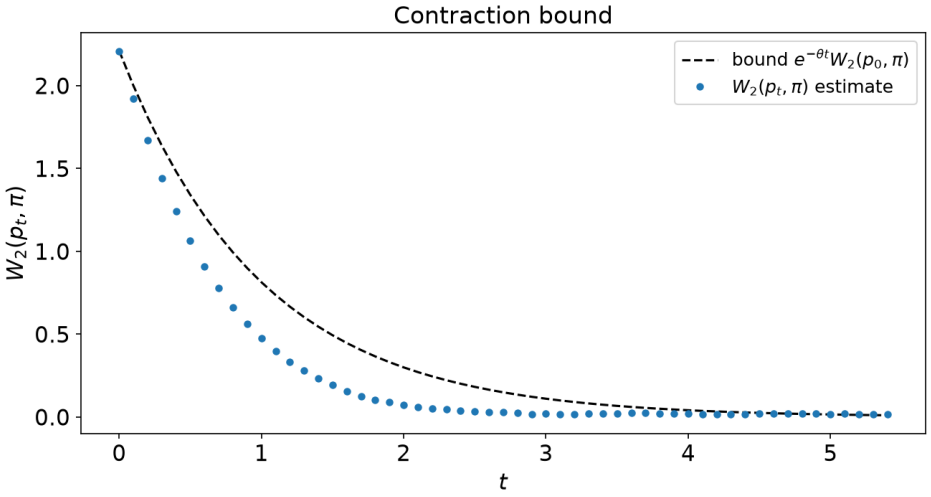


Figure 6: Contraction bound (dashed) and estimated $W_2(p_t, \pi)$ (dots), $\theta = 1$.

t	estimate	bound	bound / estimate
0.0	2.208	2.211	1.00
0.9	0.562	0.899	1.60
1.8	0.105	0.365	3.49
2.7	0.028	0.149	5.31
3.6	0.025	0.060	2.41
4.5	0.020	0.025	1.20
5.4	0.017	0.010	0.58

Table 2: Estimated $W_2(p_t, \pi)$ against the contraction bound.

How tight is the bound? At $t = 0$ the bound equals the true distance by construction. After that the estimate falls faster than the bound, and the gap widens: the bound is 1.6 times the estimate at $t = 0.9$ and over 5 times at $t = 2.7$. So the bound has the right form (exponential decay) but is loose for this p_0. The reason is that the coupling argument in 2(c) only uses how fast two particles approach each other, which is rate θ. For our symmetric p_0 the mean is already 0 (1(a)), so the actual distance to π shrinks faster than this worst case.

After $t \approx 3$ the ratio falls again and drops below 1 near T, i.e. the estimate appears to break the bound. This is not a real violation. The true $W_2(p_t, \pi)$ keeps decreasing, but the estimate cannot go below its noise floor of about 0.02 (see 2(a)), while the bound continues down to 0.01. Checking the bound at such small distances would require many more samples.

2(e) Collapse against θt

We repeat 2(a) for $\theta \in \{0.5, 1, 2\}$, sampling each run at the same values of θt (so at $t = \theta t/\theta$). Figure 7 plots all six curves against θt. For each distance, the three values of θ fall on the same curve: at any θt they differ by at most 0.016 for $W_2(p_t, \pi)$ and 0.013 for $W_2(p_t, p_0)$, which is the size of the sampling noise. This confirms once more that the law of X_t depends on θ and t only through θt (1(d)), so every distance computed from it does too.

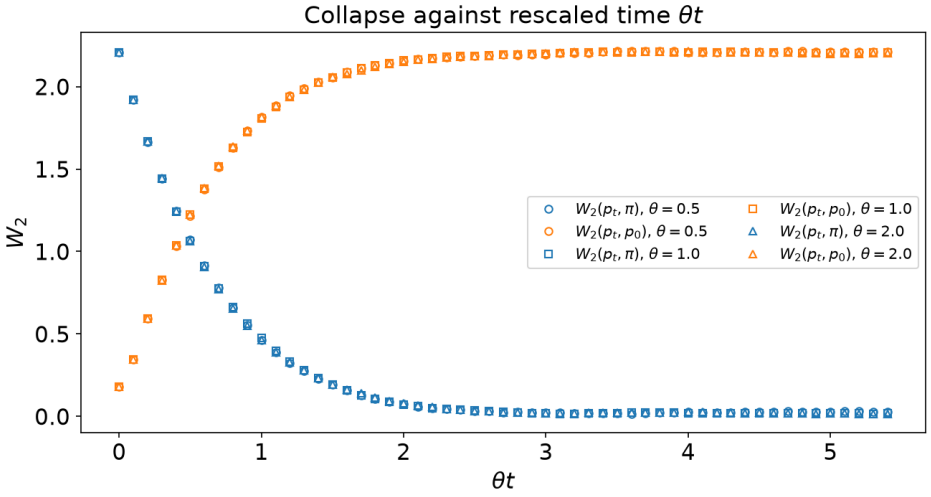


Figure 7: $W_2(p_t, \pi)$ (blue) and $W_2(p_t, p_0)$ (orange) for $\theta \in \{0.5, 1, 2\}$ against rescaled time θt.

Part 3: Score Function and the Stein Identity

The score of p_t is $s_t(x) = \partial_x \log p_t(x)$.

Derivations

3(a) Fokker–Planck equation

For an SDE $\mathrm{d}X_t = f(X_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t$, the density p_t of X_t evolves by the Fokker–Planck equation

$$\partial_t p_t(x) = -\partial_x\big(f(x,t)\,p_t(x)\big) + \tfrac{1}{2}g(t)^2\,\partial_x^2 p_t(x).$$

For (1), $f(x,t) = -\theta x$ and $g(t) = \sqrt{2\theta}$, so $\frac{1}{2}g^2 = \theta$ and

$$\partial_t p_t(x) = -\partial_x\big(-\theta x\,p_t(x)\big) + \theta\,\partial_x^2 p_t(x) = \theta\big(\partial_x(x\,p_t(x)) + \partial_x^2 p_t(x)\big).$$

Bringing the right-hand side to the left,

$$\partial_t p_t + \partial_x\Big(-\theta\big(x\,p_t + \partial_x p_t\big)\Big) = 0,$$

which is the flux form $\partial_t p_t + \partial_x J_t = 0$ with

$$\boxed{J_t(x) = -\theta\big(x\,p_t(x) + \partial_x p_t(x)\big).}$$

Stationary flux. For $\pi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$,

$$\partial_x \pi(x) = \frac{1}{\sqrt{2\pi}}\big(-x\,e^{-x^2/2}\big) = -x\,\pi(x),$$

so

$$J_\pi(x) = -\theta\big(x\,\pi(x) + \partial_x\pi(x)\big) = -\theta\big(x\,\pi(x) - x\,\pi(x)\big) = 0$$

for every x. The flux vanishes identically, so $\partial_t \pi = -\partial_x J_\pi = 0$ and π is stationary, in agreement with 1(a). Note that $J_t = -\theta\,p_t\,(x + s_t(x))$, so $J_t \equiv 0$ exactly when $s_t(x) = -x$, which is the score of π.

3(b) Score in closed form

From 1(c), $p_t(x) = \frac{1}{2}\varphi_1(x) + \frac{1}{2}\varphi_2(x)$ with

$$\varphi_1(x) = \varphi(x \mid -m_t, v_t), \qquad \varphi_2(x) = \varphi(x \mid m_t, v_t), \qquad m_t = 3e^{-\theta t}, \quad v_t = 1 - 0.75e^{-2\theta t}.$$

By the chain rule, $s_t(x) = \partial_x \log p_t(x) = \partial_x p_t(x)/p_t(x)$, and $\partial_x p_t = \frac{1}{2}\partial_x\varphi_1 + \frac{1}{2}\partial_x\varphi_2$. For a Gaussian density $\varphi(x \mid m, v) = \frac{1}{\sqrt{2\pi v}}\exp\big(-\frac{(x-m)^2}{2v}\big)$,

$$\partial_x\varphi(x \mid m, v) = \frac{1}{\sqrt{2\pi v}}\Big(-\frac{x-m}{v}\Big)\exp\Big(-\frac{(x-m)^2}{2v}\Big) = -\frac{x-m}{v}\,\varphi(x \mid m, v).$$

Hence $\partial_x\varphi_k = s_{k,t}\,\varphi_k$ with component scores

$$s_{1,t}(x) = -\frac{x + m_t}{v_t}, \qquad s_{2,t}(x) = -\frac{x - m_t}{v_t}.$$

Substituting,

$$s_t(x) = \frac{\frac{1}{2}s_{1,t}\varphi_1 + \frac{1}{2}s_{2,t}\varphi_2}{\frac{1}{2}\varphi_1 + \frac{1}{2}\varphi_2} = w_{1,t}(x)\,s_{1,t}(x) + w_{2,t}(x)\,s_{2,t}(x),$$

with

$$w_{1,t}(x) = \frac{\varphi_1(x)}{\varphi_1(x) + \varphi_2(x)}, \qquad w_{2,t}(x) = \frac{\varphi_2(x)}{\varphi_1(x) + \varphi_2(x)}.$$

The weights are non-negative and sum to one, so s_t is a weighted average of the two component scores. By Bayes' rule, with the label Z from 1(c),

$$w_{k,t}(x) = \frac{P(Z=k)\,p(x \mid Z=k)}{p_t(x)} = P(Z=k \mid X_t = x),$$

the posterior probability that a particle found at x came from component k. Therefore

$$\boxed{s_t(x) = w_{1,t}(x)\,s_{1,t}(x) + w_{2,t}(x)\,s_{2,t}(x).}$$

As $t \to \infty$, $m_t \to 0$ and $v_t \to 1$, so both component scores tend to $-x$, and hence so does $s_t(x)$: the score of π.

3(c) The score stays finite

Taken literally, $\sup_x |s_t(x)|$ is infinite for every t: the score grows linearly in $|x|$, and even the limiting score $-x$ is unbounded. What does stay finite is the score at every point and on every bounded set of x, uniformly over all $t \geq 0$, including $t = 0$. This is the property we prove.

By 3(b), $s_t(x)$ is a weighted average of $s_{1,t}(x)$ and $s_{2,t}(x)$ with weights in $[0,1]$ that sum to one. A weighted average is never larger in size than the larger of the two values being averaged, so

$$|s_t(x)| \leq \max\big(|s_{1,t}(x)|,\, |s_{2,t}(x)|\big) = \max\Big(\frac{|x+m_t|}{v_t},\, \frac{|x-m_t|}{v_t}\Big) \leq \frac{|x|+m_t}{v_t}.$$

For all $t \geq 0$ we have $m_t \leq 3$ and $v_t = 1 - 0.75e^{-2\theta t} \geq 0.25$, so

$$|s_t(x)| \leq 4|x| + 12 \qquad \text{for all } x \in \mathbb{R},\ t \geq 0.$$

Hence $\sup_{|x|\leq L} |s_t(x)| \leq 4L + 12$ for every L, with a bound that does not depend on t.

Which feature of p_0 guarantees this. The key quantity is the lower bound $v_t \geq 0.25$, which comes from the components of p_0 having positive variance 0.5^2. The variance of each component starts at 0.25 and only grows, so the $1/v_t$ factors in the score never blow up.

Two point masses. If $p_0 = \frac{1}{2}\delta_{-3} + \frac{1}{2}\delta_3$, the same computation as in 1(c) gives components $\mathcal{N}(\mp 3e^{-\theta t}, v_t)$ with $v_t = 1 - e^{-2\theta t}$, so $v_t \to 0$ as $t \to 0$.

- At $t = 0$, p_0 has no density, so $\log p_0$ and the score are not defined at all.
- For small $t > 0$, $v_t \approx 2\theta t$, so both component scores $-(x \pm m_t)/v_t$ are huge. Take for example $x = 1$. It is much closer to the right component, so $w_{2,t}(1) \to 1$ and $s_t(1) \approx s_{2,t}(1) = (m_t - 1)/v_t \approx 2/(2\theta t) \to \infty$. The same happens at every $x \neq \pm 3$.

So with point masses the score is finite for each $t > 0$ but not uniformly in t: it blows up like $1/t$ as $t \to 0$. This matters for diffusion models, where the score must be learned and evaluated down to small t.

Implementation

Density and score over time

Figure 8 plots p_t and the closed-form score s_t from 3(b) on a common grid at $t \in \{0, 0.5, 1, 2, T\}$.

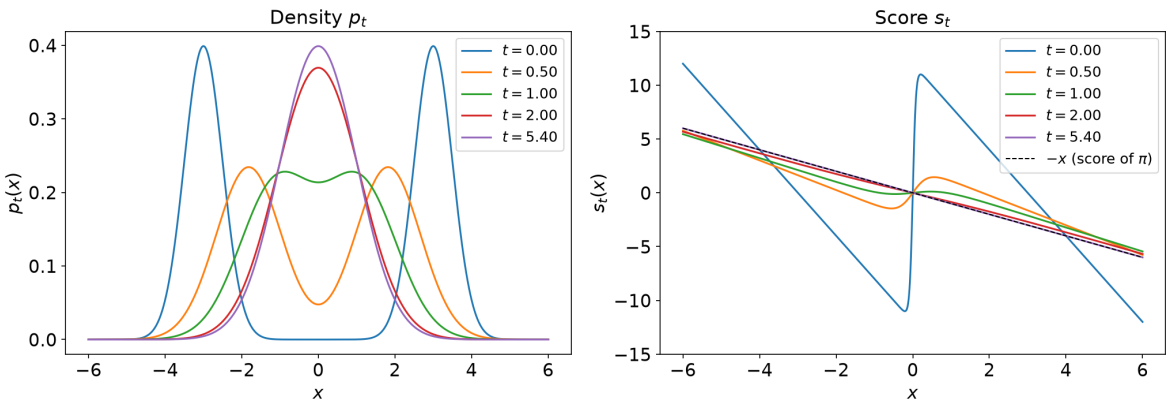


Figure 8: Density p_t (left) and score s_t (right) at several times; the dashed line is $-x$, the score of π.

The score points uphill on the density: it is positive where p_t increases and negative where it decreases, and zero at every peak and trough.

- $t = 0$**.** Inside each bump the score is close to that bump's own score, a steep line $-(x \pm 3)/0.25$ through the bump's centre. Near $x = 0$ the weights in 3(b) switch from one component to the other, so the score jumps sharply from about -11 to $+11$. This S-shaped jump marks the trough between the two modes.
- $t = 0.5$ **and** $t = 1$**.** The components move inwards and widen, so the lines become less steep and the jump at the origin becomes smaller and smoother. While the density is bimodal, the score still crosses zero three times (at the two peaks and at the trough).
- **After** $t^* \approx 1.14$**.** The density has a single peak, so the score crosses zero only once, at $x = 0$. By $t = 2$ it is already close to a straight line.
- **Limit.** As $t \to \infty$, $m_t \to 0$ and $v_t \to 1$, so the score tends to

$$s_\infty(x) = -x,$$

the score of $\pi = \mathcal{N}(0, 1)$. At $t = T$ it is indistinguishable from $-x$ on the plot.

Stein identity

Statement and proof. Let f be differentiable with

$$f(x)\, p_t(x) \to 0 \text{ as } |x| \to \infty, \qquad \mathbb{E}_{p_t}|f'(X)| < \infty, \qquad \mathbb{E}_{p_t}|f(X)\, s_t(X)| < \infty.$$

Since $s_t = \partial_x p_t / p_t$, we have $f\, s_t\, p_t = f\, \partial_x p_t$. Integrating by parts,

$$\mathbb{E}_{p_t}[f'(X)] = \int_{-\infty}^{\infty} f'(x)\, p_t(x)\, \mathrm{d}x = \Big[f(x)\, p_t(x)\Big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f(x)\, \partial_x p_t(x)\, \mathrm{d}x = -\mathbb{E}_{p_t}[f(X)\, s_t(X)],$$

where the boundary term vanishes by the decay condition. Because p_t has Gaussian tails, the condition holds for any f that grows at most polynomially, which includes all four test functions.

Numerical check. We use the $N = 2 \times 10^4$ particles of the forward simulation at $t \in \{0, 1.35, 2.70, 4.05, 5.40\}$ and evaluate the exact score s_t from 3(b) at each particle. For each test function we report the Stein residual

$$R_f = \Big|\frac{1}{N}\sum_{i=1}^{N} f'(X_i) + \frac{1}{N}\sum_{i=1}^{N} f(X_i)\, s_t(X_i)\Big|$$

and its standard error SE, the sample standard deviation of $f'(X_i) + f(X_i)s_t(X_i)$ divided by $\sqrt{N}$. A check passes when $R_f \le 2\,\mathrm{SE}$.

	$f = x$		$f = x^2$		$f = \sin x$		$f = e^{-x^2/2}$	
t	R_f	SE	R_f	SE	R_f	SE	R_f	SE
0.00	0.0565	0.0431	0.0712	0.1422	0.0078	0.0090	0.0001	0.0010
1.35	0.0019	0.0111	0.0194	0.0259	0.0044	0.0067	0.0027	0.0047
2.70	0.0078	0.0102	0.0041	0.0203	0.0006	0.0070	0.0052	0.0061
4.05	0.0020	0.0100	0.0259	0.0186	0.0048	0.0071	0.0000	0.0062
5.40	0.0057	0.0102	0.0244	0.0191	0.0029	0.0071	0.0018	0.0062

Table 3: Stein residuals R_f and standard errors. Every R_f is below $2\,\mathrm{SE}$; the largest ratio is $R_f/\mathrm{SE} = 1.39$.

All 20 checks pass. The ratio R_f/SE ranges from 0.00 to 1.39, which is what we expect for a quantity whose true mean is zero. The standard errors are largest at $t = 0$, where the score is steep ($|s_0|$ reaches about 11), so the terms $f(X_i)s_t(X_i)$ vary the most.

Part 4: Reverse Diffusion with the Exact Score

For a forward SDE $\mathrm{d}X_t = f(X_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t$ with marginals p_t, Anderson's time reversal states that the reverse-time process satisfies

$$\mathrm{d}X_t = \big[f(X_t, t) - g(t)^2 s_t(X_t)\big]\mathrm{d}t + g(t)\,\mathrm{d}\bar{W}_t, \tag{4}$$

where $\bar{W}$ is a Brownian motion running backwards in time and $\mathrm{d}t < 0$.

Derivations

4(a) Reverse SDE for the OU process

For (1), $f(x, t) = -\theta x$ and $g(t) = \sqrt{2\theta}$, so $g^2 = 2\theta$. Substituting into (4),

$$\mathrm{d}X_t = \big[-\theta X_t - 2\theta\, s_t(X_t)\big]\mathrm{d}t + \sqrt{2\theta}\,\mathrm{d}\bar{W}_t = -\theta\big[X_t + 2s_t(X_t)\big]\mathrm{d}t + \sqrt{2\theta}\,\mathrm{d}\bar{W}_t.$$

The reverse drift is

$$\boxed{f_{\mathrm{rev}}(x, t) = -\theta\big[x + 2s_t(x)\big].}$$

4(b) Discrete reverse update

We step from time t down to $t - \Delta t$ with $\Delta t > 0$, so

$$\mathrm{d}X_t \approx X_{t-\Delta t} - X_t, \qquad \mathrm{d}t = -\Delta t, \qquad \mathrm{d}\bar{W}_t \approx \bar{W}_{t-\Delta t} - \bar{W}_t \sim \mathcal{N}(0, \Delta t).$$

The Brownian increment has variance equal to the length of the time step, so we write $\mathrm{d}\bar{W}_t = \sqrt{\Delta t}\,Z$ with $Z \sim \mathcal{N}(0, 1)$; its sign does not matter because Z is symmetric. Substituting into 4(a),

$$X_{t-\Delta t} - X_t = -\theta\big[X_t + 2s_t(X_t)\big](-\Delta t) + \sqrt{2\theta\Delta t}\,Z,$$

which gives

$$\boxed{X_{t-\Delta t} = X_t + \big[\theta X_t + 2\theta\, s_t(X_t)\big]\Delta t + \sqrt{2\theta\Delta t}\,Z.} \tag{5}$$

The negative $\mathrm{d}t$ flips the sign of *both* drift terms.

Note on the sign in the assignment. The assignment's equation (9) has $-2\theta s_t(X_t)$ in place of $+2\theta s_t(X_t)$. That version is incorrect. To see this, take $p_t = \pi$, so $s_t(x) = -x$; the reverse process must then leave π unchanged.

- With (5): $X_{t-\Delta t} = (1-\theta\Delta t)X_t + \sqrt{2\theta\Delta t}\, Z$, whose variance is $(1-\theta\Delta t)^2 + 2\theta\Delta t = 1+\theta^2\Delta t^2$. So π is preserved up to $O(\Delta t^2)$.
- With the printed minus sign: $X_{t-\Delta t} = (1+3\theta\Delta t)X_t + \sqrt{2\theta\Delta t}\, Z$. Every step multiplies by more than 1, and the samples blow up.

All results below use (5).

4(c) Stability near a mode

Near one mode of p_0 the score is approximately that of a single Gaussian, $s_t(x) \approx -(x-m)/v$, where m is the centre and v the variance of the mode. Write the deviation from the mode as $y_t = X_t - m$, so $X_t = y_t + m$. Substituting into (5),

$$y_{t-\Delta t} + m = (y_t + m) + \Big[\theta(y_t + m) - 2\theta\,\frac{y_t}{v}\Big]\Delta t + \sqrt{2\theta\Delta t}\, Z,$$

so

$$y_{t-\Delta t} = \underbrace{\Big[1 - \theta\Big(\frac{2}{v} - 1\Big)\Delta t\Big]}_{A}\, y_t + \theta m\, \Delta t + \sqrt{2\theta\Delta t}\, Z.$$

The term $\theta m\Delta t$ is a constant shift and the noise term does not depend on y_t, so neither affects whether deviations grow. The recursion does not amplify y_t when $|A| \le 1$, i.e. $-1 \le A \le 1$.

- **Upper inequality** $A \le 1$: $\theta\big(\frac{2}{v} - 1\big)\Delta t \ge 0$, which holds for every $\Delta t > 0$ when $v < 2$.
- **Lower inequality** $A \ge -1$: $\theta\big(\frac{2}{v} - 1\big)\Delta t \le 2$, i.e.

$$\boxed{\Delta t \le \frac{2}{\theta\big(\frac{2}{v} - 1\big)}.}$$

At $v = v_0 = 0.25$ and $\theta = 1$, $\frac{2}{v} = 8$ and

$$\Delta t \le \frac{2}{8-1} = \frac{2}{7} \approx 0.286.$$

The narrowest modes (smallest v) occur at $t = 0$, so this is the most restrictive value. The default $\Delta t = 10^{-3}$ is far below it.

Implementation

4(a) and 4(c) Running the reverse SDE

We implement (5) with the exact score from 3(b), stepping $t = T, T-\Delta t, \ldots, \Delta t$ with $\Delta t = 10^{-3}$ and $T = 5.40$, and evaluating the score at the current time t in each step. The $N = 2 \times 10^4$ particles at the end form the empirical law $\hat{p}_0$.

4(b) Starting from π instead of p_T

We start the reverse process from N samples of $\pi = \mathcal{N}(0,1)$ rather than from p_T. This is justified by Part 2: the contraction bound gives $W_2(p_T, \pi) \le e^{-\theta T} W_2(p_0, \pi)$, and T was chosen in 2(d) so that this is at most 10^{-2}. So the starting distribution differs from the correct one, p_T, by at most 0.01 in W_2. This is the error the substitution introduces.

4(d) Comparing $\hat{p}_0$ with p_0

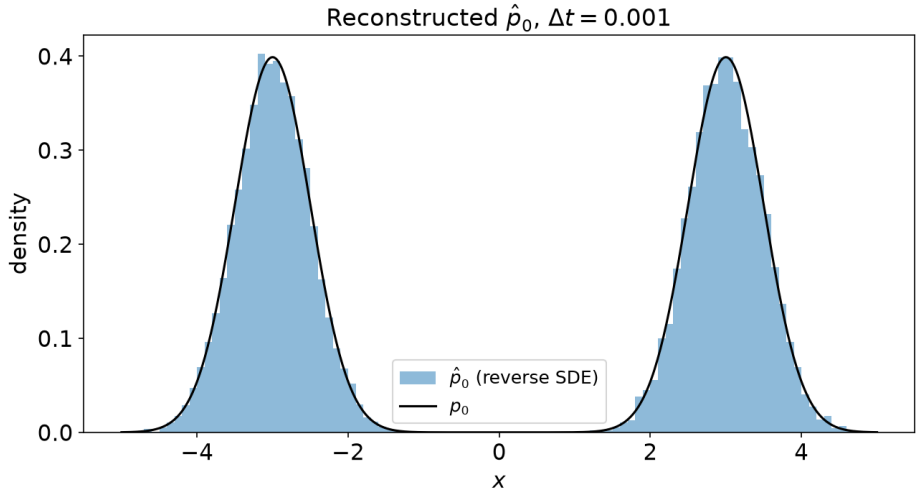


Figure 9: Histogram of $\hat{p}_0$ from the reverse SDE ($\Delta t = 10^{-3}$) against the true p_0.

Figure 9 shows that the reverse SDE recovers both modes, with the right locations, heights and widths. Starting from a single Gaussian, the reverse dynamics separates the particles into the two original modes.

	$\hat{p}_0$	p_0
mean	0.005	0
variance	9.255	9.25

Table 4: First two moments. The true variance is $3^2 + 0.5^2 = 9.25$.

Both moments agree with p_0 to within sampling error (the standard errors are about 0.02 for the mean and 0.02 for the variance). The estimated distance is $W_2(\hat{p}_0, p_0) = 0.188$. For comparison, two independent samples of size N drawn directly from p_0 are 0.227 apart under the same estimator, so $\hat{p}_0$ is as close to p_0 as a sample from p_0 itself. These values are large because of how the estimator handles the two modes: the number of points in each mode varies randomly by about $\sqrt{N}/2 \approx 70$ between samples, and after sorting, these surplus points are paired with points in the other mode, about 6 away.

4(e) Convergence in Δt

Δt	$W_2(\hat{p}_0, p_0)$
10^{-3}	0.188
5×10^{-4}	0.315
2.5×10^{-4}	0.149

Table 5: $W_2(\hat{p}_0, p_0)$ for the reverse SDE as Δt is halved and quartered.

The values do not decrease with Δt: the ratios give observed orders $\log_2(0.188/0.315) = -0.74$ and $\log_2(0.315/0.149) = 1.08$, with no consistent pattern. So no order of convergence can be identified at these step sizes. The reason is that all three values are of the same size as the W_2 distance between two independent samples of p_0 (0.227, see 4(d)). The discretisation error of

Euler–Maruyama at $\Delta t \leq 10^{-3}$ is much smaller than this sampling noise, so it cannot be seen. The theoretical expectation is weak order 1, i.e. an error proportional to Δt.

4(f) Probability flow ODE

The probability flow ODE $\mathrm{d}X_t/\mathrm{d}t = -\theta X_t - \theta s_t(X_t)$ (equation (10) of the assignment) is discretised with the explicit Euler method, again with $\mathrm{d}t = -\Delta t$:

$$X_{t-\Delta t} = X_t + \theta\big[X_t + s_t(X_t)\big]\Delta t.$$

It starts from the same N samples of π as the SDE and has no noise term, so each particle follows a deterministic path. Figure 10 compares $W_2(\hat{p}_0, p_0)$ for the two schemes at equal Δt, from 10^{-3} up to the coarse step 0.2 (still below the stability limit $2/7$ from 4(c)).

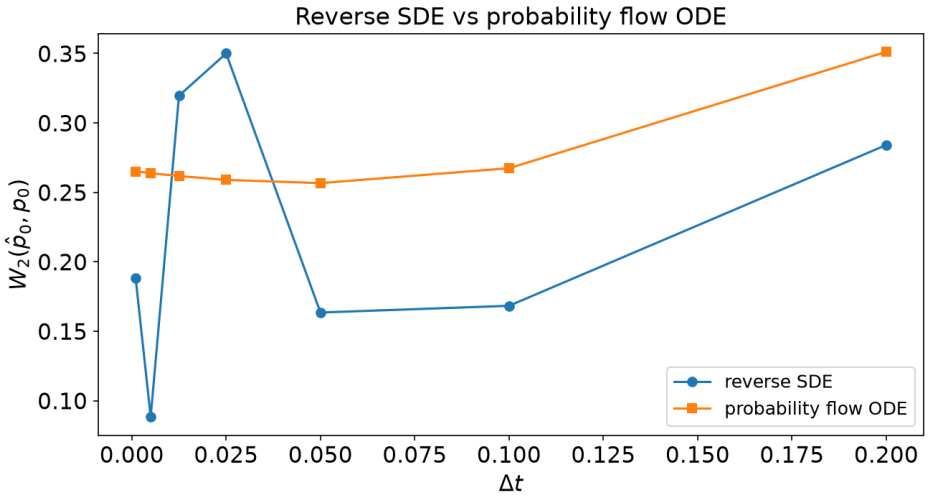


Figure 10: $W_2(\hat{p}_0, p_0)$ against Δt for the reverse SDE and the probability flow ODE.

- **ODE.** The curve is smooth: since there is no noise and every run starts from the same particles, the only thing that changes with Δt is the discretisation error. For $\Delta t \leq 0.05$ it is flat at about 0.26, the sampling floor of this particular set of particles. At $\Delta t = 0.1$ and 0.2 it rises to 0.27 and 0.35, so the discretisation error becomes visible at coarse steps.
- **SDE.** Each run uses new noise, so the number of particles that end up in each mode changes from run to run, and the curve jumps between 0.09 and 0.35 with no trend in Δt. Even at $\Delta t = 0.2$, its value (0.28) lies within this run-to-run range.

At the coarsest step, $\Delta t = 0.2$, the SDE gives the smaller distance (0.28 against 0.35). But since the SDE values vary by about ± 0.1 between runs, a single run per step size cannot show reliably which scheme is more accurate at coarse steps. What the plot does show is that the ODE's error grows clearly once Δt exceeds about 0.05, while for $\Delta t \leq 0.05$ both schemes are limited by sampling noise rather than by the step size.

Part 5: Effect of Imperfect Score Estimation

We repeat the Part 4 reconstruction (same N, Δt, T, seed, start from π) with an estimated score $\hat{s}_t$ in (5):

$$X_{t-\Delta t} = X_t + \big[\theta X_t + 2\theta\, \hat{s}_t(X_t)\big]\Delta t + \sqrt{2\theta\Delta t}\, Z.$$

For each run we record $W_2(\hat{p}_0, p_0)$ and the Fisher divergence $D_F = \frac{1}{T}\int_0^T \mathbb{E}_{p_t}\big[(\hat{s}_t(X) - s_t(X))^2\big]\mathrm{d}t$, where $\mathbb{E}_{p_t}$ is the average over the Part 1 forward particles at 55 equally spaced times and the time integral uses the trapezoidal rule. As in 4(d), $W_2 \approx 0.2$ is the noise floor of the estimator.

Case A: Random error

On a grid of $K = 200$ points in $[-6, 6]$ we draw $e_k \sim \mathcal{N}(0, \sigma_\epsilon^2)$ once, and set $\hat{s}_t = s_t + e$, with e the linear interpolation of the e_k, held fixed for the whole run.

σ_ϵ	$W_2(\hat{p}_0, p_0)$	D_F	mean	variance
0.1	0.315	0.005	−0.030	9.244
0.3	0.445	0.047	−0.084	9.212
1.0	0.875	0.527	−0.304	9.054
3.0	1.649	4.740	−0.872	8.316

Table 6: Case A. The true mean and variance are 0 and 9.25.

W_2 grows with σ_ϵ, and D_F grows like σ_ϵ^2. A single fixed draw of e is not zero on average where the particles are, so it acts locally like a bias; in this draw it shifts the mean to the left.

Case B: Systematic bias

Here $\hat{s}_t = s_t + c$, so $D_F = c^2$ exactly.

Prediction. Take a single Gaussian mode, $p_t = \mathcal{N}(m_t, v_t)$ with $m_t = m_0 e^{-\theta t}$ and $v_t = 1 - (1 - v_0)e^{-2\theta t}$, so $s_t(x) = -(x - m_t)/v_t$.

1. Substituting $\hat{s}_t = s_t + c$ into the update and taking expectations, with $\mu_t = \mathbb{E}[X_t]$,

$$\mu_{t-\Delta t} = \mu_t + \theta\mu_t\Delta t + 2\theta\,\mathbb{E}[s_t(X_t)]\,\Delta t + 2\theta c\,\Delta t, \qquad \mathbb{E}[s_t(X_t)] = -\frac{\mu_t - m_t}{v_t}.$$

   This expectation is 0 only when $\mu_t = m_t$, i.e. when the particles follow p_t. Once the bias has moved them, the true score pulls them back, so the term must be kept.

2. The correct mean satisfies the same recursion with $c = 0$ and $\mu_t = m_t$: $m_{t-\Delta t} = m_t + \theta m_t \Delta t$.

3. Subtracting, the shift $\delta_t = \mu_t - m_t$ satisfies

$$\delta_{t-\Delta t} = \delta_t + \theta\Big(1 - \frac{2}{v_t}\Big)\delta_t\Delta t + 2\theta c\,\Delta t.$$

4. Rearranging, $(\delta_t - \delta_{t-\Delta t})/\Delta t = \theta(2/v_t - 1)\delta_t - 2\theta c$. Letting $\Delta t \to 0$,

$$\frac{\mathrm{d}\delta}{\mathrm{d}t} = \theta\Big(\frac{2}{v_t} - 1\Big)\delta - 2\theta c, \qquad \delta_T = 0,$$

   since the run starts from $\pi \approx p_T$ at $t = T$.

5. Integrating factor $e^{-\theta t}/v_t$. Using $\mathrm{d}v_t/\mathrm{d}t = 2\theta(1 - v_t)$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(\frac{e^{-\theta t}}{v_t}\Big) = \frac{e^{-\theta t}}{v_t}\Big(-\theta - \frac{2\theta(1 - v_t)}{v_t}\Big) = -\theta\Big(\frac{2}{v_t} - 1\Big)\frac{e^{-\theta t}}{v_t},$$

   so

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(\frac{e^{-\theta t}}{v_t}\,\delta_t\Big) = \frac{e^{-\theta t}}{v_t}\Big(\frac{\mathrm{d}\delta}{\mathrm{d}t} - \theta\Big(\frac{2}{v_t} - 1\Big)\delta_t\Big) = -2\theta c\,\frac{e^{-\theta t}}{v_t}.$$

6. Integrating from 0 to T and using $\delta_T = 0$: $-\delta_0/v_0 = -2\theta c \int_0^T e^{-\theta s}/v_s \, \mathrm{d}s$, i.e.

$$\boxed{\delta_0 = 2\theta c\, v_0 \int_0^T \frac{e^{-\theta s}}{v_s}\, \mathrm{d}s \approx 0.76\, c} \qquad (v_0 = 0.25,\ \theta = 1,\ T = 5.40).$$

A constant bias shifts the mean of the reconstruction in the direction of c, by an amount proportional to c.

c	$W_2(\hat{p}_0, p_0)$	D_F	mean	variance
-1.0	4.540	1.00	-3.692	0.534
-0.5	3.637	0.25	-2.673	3.901
-0.2	2.235	0.04	-1.252	8.013
0.2	2.186	0.04	1.265	7.974
0.5	3.602	0.25	2.692	3.806
1.0	4.497	1.00	3.700	0.499

Table 7: Case B.

Test. The direction is confirmed: the mean moves with the sign of c, symmetrically. The size is much larger than $0.76\, c$ (e.g. 1.26 at $c = 0.2$) and not linear in c. The prediction covers a single mode; with two modes the bias also moves particles from one mode to the other. A mean of 1.26 corresponds to about 71% of the particles in the right mode, and at $|c| = 1$ almost all particles end up in one mode, which is why the variance drops to about 0.5.

Case C: Multiplicative error

Here $\hat{s}_t = (1+\alpha)s_t$, so $D_F = \alpha^2 \frac{1}{T} \int_0^T \mathbb{E}_{p_t}[s_t^2]\, \mathrm{d}t \approx \alpha^2$.

Prediction. Take the same single Gaussian mode, so $\hat{s}_t(x) = -(1+\alpha)(x - m_t)/v_t$.

1. Let $y_t = X_t - m_t$. Substituting into the update and using $m_{t-\Delta t} = m_t + \theta m_t \Delta t$ (step 2 of Case B), the m_t terms cancel:

$$y_{t-\Delta t} = (1 + a_t \Delta t)\, y_t + \sqrt{2\theta \Delta t}\, Z, \qquad a_t = \theta\Big(1 - \frac{2(1+\alpha)}{v_t}\Big).$$

2. The mean of y starts at 0 and stays 0. For the variance $w_t = \mathbb{E}[y_t^2]$, using that Z is independent of y_t with $\mathbb{E}Z^2 = 1$,

$$w_{t-\Delta t} = (1 + a_t \Delta t)^2 w_t + 2\theta \Delta t = w_t + 2a_t w_t \Delta t + 2\theta \Delta t + O(\Delta t^2).$$

3. Letting $\Delta t \to 0$,

$$\frac{\mathrm{d}w}{\mathrm{d}t} = -2a_t w - 2\theta = 2\theta\Big(\frac{2(1+\alpha)}{v_t} - 1\Big) w - 2\theta, \qquad w_T = 1.$$

4. Try $w_t = v_t/(1+2\alpha)$. With $\mathrm{d}v_t/\mathrm{d}t = 2\theta(1 - v_t)$:

$$\text{left} = \frac{2\theta(1-v_t)}{1+2\alpha}, \qquad \text{right} = \frac{2\theta\big(2(1+\alpha) - v_t\big) - 2\theta(1+2\alpha)}{1+2\alpha} = \frac{2\theta(1-v_t)}{1+2\alpha}.$$

So it is a solution.

5. The difference $u_t = w_t - v_t/(1+2\alpha)$ solves the equation without the -2θ term, so $u_0 = u_T \exp\big(-\int_0^T 2\theta\,(2(1+\alpha)/v_s - 1)\,\mathrm{d}s\big)$. Since $v_s \le 1$, the exponent is at most $-2\theta(1+2\alpha)T$, so for $\alpha > -\frac{1}{2}$ the difference is negligible after a long run:

$$\boxed{\mathrm{Var}(\hat{p}_0) \approx \frac{v_0}{1+2\alpha}, \qquad \alpha > -\tfrac{1}{2}.}$$

A score that is too strong ($\alpha > 0$) shrinks the variance; one that is too weak ($\alpha < 0$) inflates it, and for $\alpha \le -\frac{1}{2}$ the variance is not controlled.

α	$W_2(\hat{p}_0, p_0)$	D_F	mean	variance	predicted $9 + 0.25/(1+2\alpha)$
-0.50	1.344	0.250	0.028	18.004	–
-0.25	0.216	0.063	0.021	10.090	9.50
-0.10	0.177	0.010	0.008	9.442	9.31
0.10	0.203	0.010	0.004	9.162	9.21
0.25	0.260	0.063	-0.003	9.091	9.17
0.50	0.257	0.250	0.003	9.062	9.13

Table 8: Case C. For two modes at ± 3, the total variance is about 9 plus the within-mode variance $0.25/(1+2\alpha)$.

Test. The mean stays at 0. The variance moves in the predicted direction for every α: above 9.25 for $\alpha < 0$ and below for $\alpha > 0$. The size is close for $\alpha > 0$ and underestimated for $\alpha < 0$, where a weaker score also separates the two modes less cleanly; at $\alpha = -0.5$ the variance reaches 18.0.

Comparing the cases

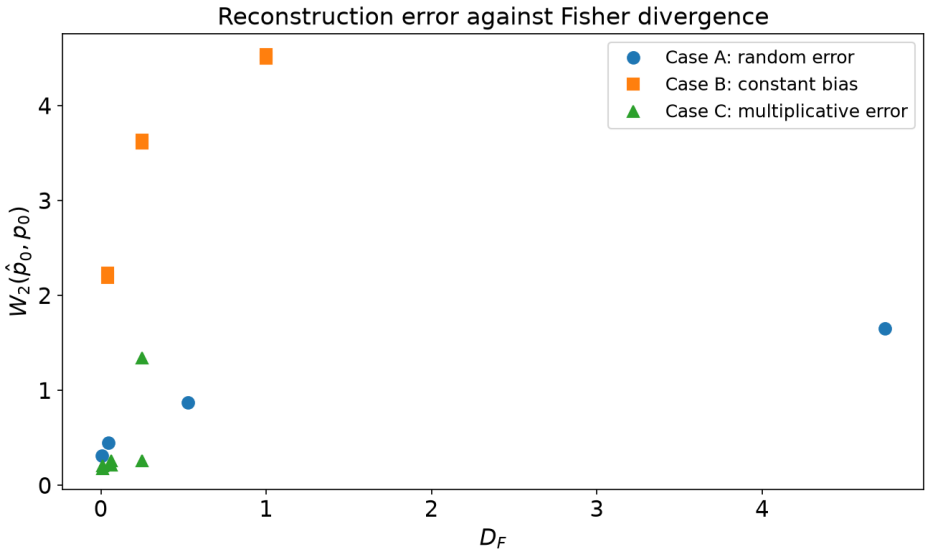


Figure 11: $W_2(\hat{p}_0, p_0)$ against D_F for all runs.

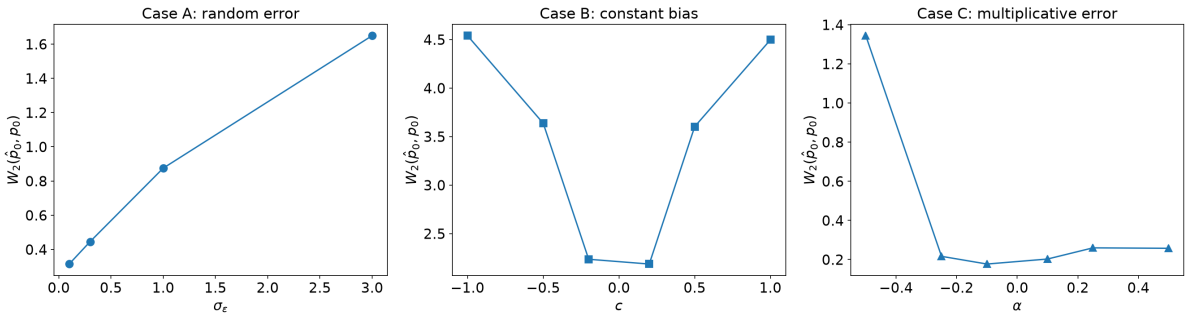


Figure 12: $W_2(\hat{p}_0, p_0)$ against the native error parameter of each case.

D_F alone does not determine the reconstruction error. At similar D_F, the bias is by far the most harmful ($W_2 \approx 2.2$ at $D_F = 0.04$), against 0.45 for the random error ($D_F = 0.047$) and about 0.2 for the multiplicative error ($D_F = 0.063$). A bias pushes every particle the same way and moves probability between the modes, 6 apart. A multiplicative error keeps the drift symmetric, so only the spread of each mode changes. The random error lies in between.

Part 6: Application Challenge, Noisy Sensor Measurements

We observe $Y^{(i)} = X_0^{(i)} + \eta^{(i)}$, $i = 1, \dots, N$, with $X_0^{(i)} \sim p_0$ and $\eta^{(i)} \sim \mathcal{N}(0, \sigma^2)$ independent, where $\sigma = \sigma_{\text{sensor}}$. The target is p_0.

Derivations

6(a) Rescaling to a forward marginal

By 1(b), $X_t = e^{-\theta t} X_0 + \sqrt{1 - e^{-2\theta t}}\, Z$ with $Z \sim \mathcal{N}(0, 1)$ independent of X_0. For $a > 0$,

$$aY = aX_0 + a\eta, \qquad a\eta \sim \mathcal{N}(0, a^2\sigma^2) \text{ independent of } X_0.$$

Both are "X_0 scaled, plus independent Gaussian noise", so aY has the law p_{t^*} (for any p_0) if the scale and the noise variance match:

$$a = e^{-\theta t^*}, \qquad a^2\sigma^2 = 1 - e^{-2\theta t^*}.$$

Substituting the first into the second, $e^{-2\theta t^*}\sigma^2 = 1 - e^{-2\theta t^*}$, i.e. $e^{-2\theta t^*}(1 + \sigma^2) = 1$. Hence

$$\boxed{t^* = \frac{\ln(1 + \sigma^2)}{2\theta}, \qquad a = e^{-\theta t^*} = \frac{1}{\sqrt{1 + \sigma^2}}.}$$

6(b) Sign and behaviour of t^*

For $\sigma > 0$, $1 + \sigma^2 > 1$, so $\ln(1 + \sigma^2) > 0$ and $t^* > 0$. Since $\ln(1 + \sigma^2)$ is increasing in σ, t^* grows with the sensor noise: a noisier sensor corresponds to a later, more diffused time. For small noise $t^* \approx \sigma^2/(2\theta) \to 0$, and for large noise $t^* \approx \ln \sigma/\theta \to \infty$, so t^* grows only logarithmically. At the same time $a \to 0$: the signal is scaled down more and more.

6(c) Why Y cannot be used without the factor a

The reverse process started at time t is only correct if its starting points have law p_t. Without rescaling, $Y = X_0 + \eta$ keeps the signal X_0 at full scale, while every X_t carries it scaled by $e^{-\theta t}$. For our p_0, Y is a mixture of $\mathcal{N}(\pm 3,\, 0.25 + \sigma^2)$, whereas p_t is a mixture of $\mathcal{N}(\pm 3e^{-\theta t},\, v_t)$.

Matching the mode positions requires $e^{-\theta t} = 1$, i.e. $t = 0$, but then $v_0 = 0.25 \neq 0.25 + \sigma^2$. So for $\sigma > 0$ there is no time t at which Y has law p_t. Starting the reverse process from Y at any t gives it inputs that are too wide (if t is small) or too far apart (if t is large), and the score of p_t then pushes them to the wrong places.

Task (2): Linear shrinkage λ

Treat X_0 as Gaussian with the correct first two moments, $\mathbb{E}[X_0] = 0$ and $\mathrm{Var}(X_0) = s_0^2 = 3^2 + 0.5^2 = 9.25$. For $\hat{X} = \lambda Y = \lambda X_0 + \lambda\eta$, using that X_0 and η are independent with mean 0,

$$\mathbb{E}\big[(\lambda Y - X_0)^2\big] = \mathbb{E}\big[((\lambda - 1)X_0 + \lambda\eta)^2\big] = (\lambda - 1)^2 s_0^2 + \lambda^2\sigma^2.$$

Setting the derivative in λ to zero, $2(\lambda - 1)s_0^2 + 2\lambda\sigma^2 = 0$, so

$$\boxed{\lambda = \frac{s_0^2}{s_0^2 + \sigma^2} = \frac{9.25}{9.25 + \sigma^2}.}$$

(Because the mean is 0, no intercept is needed.)

Tasks

Setup. For each $N \in \{1000, 5000, 20000\}$ and $\sigma \in \{0.1, 0.3, 1, 3\}$ (a factor of 30), we draw N observations Y and compare three reconstructions:

1. the raw observations Y;
2. the shrinkage estimate λY;
3. reverse diffusion: $X_{t^*} = aY$, integrated with (5) and the exact score from t^* down to 0.

Assumptions. For reverse diffusion we use steps of size $t^*/\lceil t^*/10^{-3}\rceil \leq 10^{-3}$, so that the run ends exactly at $t = 0$. For $\sigma = 0.1$, $t^* = 0.005$, so this is only 5 steps. Each reconstruction is compared by W_2 with a fixed reference sample of N points from p_0, using one run per (N, σ).

N	σ	t^*	raw Y	shrinkage	reverse diffusion
1000	0.1	0.005	0.075	0.073	0.071
1000	0.3	0.043	0.125	0.112	0.065
1000	1.0	0.347	0.660	0.588	0.114
1000	3.0	1.151	2.121	1.394	0.457
5000	0.1	0.005	0.051	0.053	0.050
5000	0.3	0.043	0.095	0.101	0.046
5000	1.0	0.347	0.608	0.589	0.093
5000	3.0	1.151	1.987	1.448	0.235
20000	0.1	0.005	0.175	0.175	0.176
20000	0.3	0.043	0.184	0.184	0.176
20000	1.0	0.347	0.623	0.593	0.169
20000	3.0	1.151	2.008	1.419	0.203

Table 9: $W_2(\hat{p}, p_0)$ for the three reconstructions.

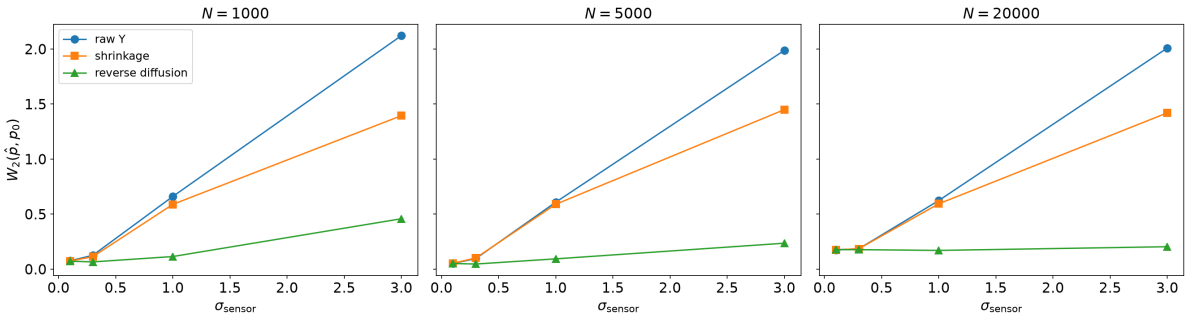


Figure 13: $W_2(\hat{p}, p_0)$ against σ_{sensor} for each N.

As in 4(d), W_2 between two independent samples of p_0 is not zero, and with two modes it varies a lot from sample to sample (0.57, 0.13 and 0.23 for the three values of N in our runs). Differences of this size are therefore not meaningful; the conclusions below rely only on larger differences.

- **Small noise** ($\sigma \leq 0.3$): all three methods are equally good, since Y is already close to X_0.
- **Raw Y:** the error grows roughly in proportion to σ, to about 2.0 at $\sigma = 3$, because the noise widens both modes and fills the gap between them.
- **Shrinkage:** better than raw Y at large σ (1.4 against 2.0 at $\sigma = 3$), but still far off. Multiplying by $\lambda < 1$ makes the spread smaller, but it also pulls the two modes towards each other and keeps them blurred. It is the best *linear* estimate of each X_0, not a reconstruction of the distribution.
- **Reverse diffusion:** stays near the noise level for all σ and N (at most 0.46, and 0.20 at $N = 20000$, $\sigma = 3$). After rescaling, the observations are exactly a sample of p_{t^*}, so the reverse process recovers p_0 up to sampling and discretisation error. Note that it reconstructs the *distribution*: each output is a new sample from p_0, not a denoised version of the corresponding $X_0^{(i)}$.